

Basic Information

Experiment No: 01

Date: 24/07/2026

Title: Student Utility Program

Objective / Aim

To design and implement a Java-based Student Utility Program demonstrating student information processing, conditional statements, looping statements, and utility methods.

System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Minimum 500 MB free storage

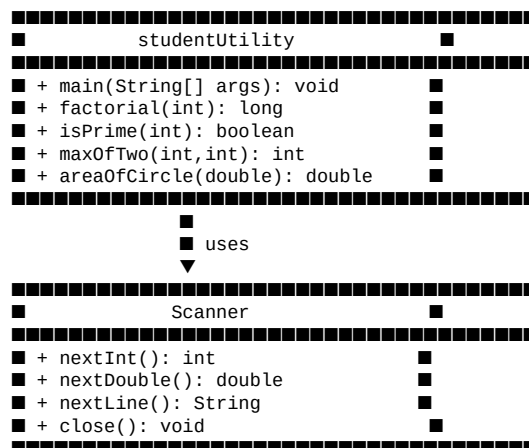
Software

- Windows / Linux / macOS
- JDK 17 or JDK 21
- Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Display the studentUtility program menu.
2. Accept the user's choice.
3. Student Information: read name, roll number, and three subject marks.
4. Conditional Statements: perform even/odd, largest number, grade, and day operations.
5. Looping Statements: generate multiplication table, numbers 1 to N, sum of N numbers, and Fibonacci series.
6. Utility Methods: calculate factorial, check prime, find maximum of two numbers, and calculate circle area.
7. Repeat the menu until the user exits.

UML Class Diagram



Main program contains menu-driven switch statements for:
Student Information, Conditional Statements, Looping Statements,
and Utility Methods.

Source Code

```
package javacore;

import java.util.Scanner;

public class studentUtility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("studentUtility program");
            System.out.println("1.Student Information");
            System.out.println("2.Conditional Statements");
            System.out.println("3.Looping Statements");
```

```

System.out.println("4.Utility methods");
System.out.println("5.Exit");
System.out.println("Enter your choice:");
choice = sc.nextInt();

switch(choice) {
case 1:
    System.out.println("Enter Name:");
    sc.nextLine();
    String name = sc.nextLine();
    System.out.println("Enter the Roll Number:");
    int roll = sc.nextInt();
    System.out.println("Enter Marks in 3 subjects");
    int m1 = sc.nextInt(), m2 = sc.nextInt(), m3 = sc.nextInt();
    int total = m1 + m2 + m3;
    double percentage = (total / 3) * 100;
    System.out.println("\n--- Student Result ---");
    System.out.println("Name: " + name);
    System.out.println("Roll No: " + roll);
    System.out.println("Total Marks: " + total);
    System.out.println("Percentage: " + percentage + "%");
    break;

case 2:
    System.out.println("\n Conditional Statement");
    System.out.println("1. Even or Odd");
    System.out.println("2. Largest of Three Numbers");
    System.out.println("3. Grade based on Percentage");
    System.out.println("4. Day of the Week");
    int condChoice = sc.nextInt();

    switch (condChoice) {
    case 1:
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        System.out.println(num % 2 == 0 ? "Even" : "Odd");
        break;
    case 2:
        System.out.print("Enter three numbers: ");
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();
        int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);
        System.out.println("Largest: " + largest);
        break;
    case 3:
        System.out.print("Enter percentage: ");
        double per = sc.nextDouble();
        if (per >= 90) System.out.println("Grade: A+");
        else if (per >= 75) System.out.println("Grade: A");
        else if (per >= 60) System.out.println("Grade: B");
        else if (per >= 40) System.out.println("Grade: C");
        else System.out.println("Grade: Fail");
        break;
    case 4:
        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();
        switch (day) {
        case 1: System.out.println("Monday"); break;
        case 2: System.out.println("Tuesday"); break;
        case 3: System.out.println("Wednesday"); break;
        case 4: System.out.println("Thursday"); break;
        case 5: System.out.println("Friday"); break;
        case 6: System.out.println("Saturday"); break;
        case 7: System.out.println("Sunday"); break;
        default: System.out.println("Invalid day!");
        }
        break;
    }
    break;

case 3:
    System.out.println("\n--- Looping Statements ---");
    System.out.println("1. Multiplication Table");
    System.out.println("2. Display numbers 1 to N");
    System.out.println("3. Sum of first N natural numbers");
    System.out.println("4. Fibonacci Series");
    int loopChoice = sc.nextInt();

    switch (loopChoice) {
    case 1:
        System.out.print("Enter number: ");
        int n = sc.nextInt();
        for (int i = 1; i <= 10; i++)
            System.out.println(n + " x " + i + " = " + (n * i));
        break;
    case 2:
        System.out.print("Enter N: ");
        int N = sc.nextInt();
        for (int i = 1; i <= N; i++) System.out.print(i + " ");
        System.out.println();
        break;
    }
}

```

```

        case 3:
            System.out.print("Enter N: ");
            int numN = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= numN; i++) sum += i;
            System.out.println("Sum = " + sum);
            break;
        case 4:
            System.out.print("Enter terms: ");
            int terms = sc.nextInt();
            int f1 = 0, f2 = 1;
            System.out.print("Fibonacci: " + f1 + " " + f2);
            for (int i = 3; i <= terms; i++) {
                int next = f1 + f2;
                System.out.print(" " + next);
                f1 = f2;
                f2 = next;
            }
            System.out.println();
            break;
    }
    break;

case 4:
    System.out.println("\n--- Methods ---");
    System.out.println("1. Factorial");
    System.out.println("2. Prime Check");
    System.out.println("3. Max of Two Numbers");
    System.out.println("4. Area of Circle");
    int methodChoice = sc.nextInt();

    switch (methodChoice) {
        case 1:
            System.out.println("Enter number: ");
            int factNum = sc.nextInt();
            System.out.println("Factorial = " + factorial(factNum));
            break;
        case 2:
            System.out.println("Enter number: ");
            int primeNum = sc.nextInt();
            System.out.println(isPrime(primeNum) ? "Prime" : "Not Prime");
            break;
        case 3:
            System.out.print("Enter two numbers: ");
            int x = sc.nextInt(), y = sc.nextInt();
            System.out.println("Max = " + maxOfTwo(x, y));
            break;
        case 4:
            System.out.print("Enter radius: ");
            double r = sc.nextDouble();
            System.out.println("Area = " + areaOfCircle(r));
            break;
    }
    break;

case 0:
    System.out.println("Exiting program...");
    break;

default:
    System.out.println("Invalid choice!");
}
} while (choice != 0);

sc.close();
}

public static long factorial(int n) {
    long fact = 1;
    for (int i = 1; i <= n; i++) fact *= i;
    return fact;
}

public static boolean isPrime(int n) {
    if (n <= 1) return false;
    for (int i = 2; i <= Math.sqrt(n); i++) {
        if (n % i == 0) return false;
    }
    return true;
}

public static int maxOfTwo(int a, int b) {
    return (a > b) ? a : b;
}

public static double areaOfCircle(double r) {
    return Math.PI * r * r;
}
}

```

Compilation & Execution Commands

```
javac studentUtility.java
java javacore.studentUtility
```

Output Screen

```
studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice: 1

Enter Name:
Ananya
Enter the Roll Number:
105
Enter Marks in 3 subjects
85 78 92

--- Student Result ---
Name: Ananya
Roll No: 105
Total Marks: 255
Percentage: 8500.0%

studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice: 3

--- Looping Statements ---
1. Multiplication Table
2. Display numbers 1 to N
3. Sum of first N natural numbers
4. Fibonacci Series
Enter your choice: 1
Enter number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70

studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice: 4

--- Methods ---
1. Factorial
2. Prime Check
3. Max of Two Numbers
4. Area of Circle
Enter your choice: 2
Enter number:
29
Prime

studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice: 0
Exiting program...
```

Result

The studentUtility program was successfully implemented in Java. It demonstrates student information processing, conditional statements, looping statements, and utility methods using a menu-driven program.